Multiplicative functions

Instructor solutions

A problem set on prime powers and divisor sums

8 problems | 100 points + 10 optional challenge points | Suggested time: 2–3 hours

Preparation. Unique factorization, gcds, finite sums, and inclusion–exclusion.

Instructions. Write on separate paper. Show factorizations and justify your answers. When asked for a proof, do not merely cite the statement. You may use results from earlier problems even if you have not proved them. Computation may check examples but does not replace a proof.

Conventions. All integers and divisors in this assignment are positive unless stated otherwise. An arithmetic function is a function on the positive integers with real values. It is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$; it is *completely multiplicative* if $f(1) = 1$ and the product rule holds for all m, n. Thus $f(\prod_i p_i^{a_i}) = \prod_i f(p_i^{a_i})$ for multiplicative f and distinct primes p_i. Empty products have value 1.

Grading note. Subpart scores total 100; the challenge is extra. Accept equivalent proofs and give method credit after a carried arithmetic error. Check uses of coprimality and the value at 1. The time estimate excludes the challenge and assumes practice writing short proofs.

I. What prime powers tell us

1. Read the definition carefully

[10 points]

Suppose f is multiplicative and

$$f(2) = 2, \qquad f(4) = 5, \qquad f(3) = -1, \qquad f(9) = 4, \qquad f(5) = 0.$$

(a) Find $f(1), f(12), f(18), f(36)$, and $f(45)$. (5)

(b) Do these data determine $f(8)$? Explain by describing two multiplicative functions satisfying all the given data but having different values at 8. (3)

(c) Could f be completely multiplicative? Justify your answer. (2)

Solution

(a) The values, in order, are

$$1, \quad f(4)f(3) = -5, \quad f(2)f(9) = 8, \quad f(4)f(9) = 20, \quad f(9)f(5) = 0.$$

The factors in each displayed product are coprime.

(b) No. Assign the five specified prime-power values as given. For a first function assign value 0 at 2^3; for a second assign value 7 there. Assign value 1 at every other unspecified prime power. Set each function's value at 1 to 1, and at a general integer take the product over its prime-power factors. Unique factorization makes both functions well defined and multiplicative, since coprime inputs have disjoint prime factors.

(c) No: complete multiplicativity would require $f(4) = f(2)^2 = 4$, but the given value is 5.

Grading: (a) One point per value. (b) Nonuniqueness: 1; two consistent prime-power assignments: 1; multiplicative extension: 1. (c) Required equality and contradiction: 2.

2. Which operations preserve multiplicativity? [10 points]

Let f and g be multiplicative functions.

(a) Prove that the pointwise product $h(n) = f(n)g(n)$ is multiplicative. Prove the same for $q(n) = f(n)/g(n)$ if $g(n) \neq 0$ for every n. (4)

(b) Explain why the pointwise sum $f + g$ is never multiplicative under our convention. (2)

(c) Normalizing the sum fixes its value at 1. Does it follow that $A(n) = (f(n) + g(n))/2$ is always multiplicative? Prove this or give an explicit counterexample with coprime inputs. (4)

Solution

(a) Both $h(1)$ and $q(1)$ are 1. For coprime m, n,

$$h(mn) = f(m)f(n)g(m)g(n) = h(m)h(n), \qquad q(mn) = \frac{f(m)f(n)}{g(m)g(n)} = q(m)q(n).$$

Nonvanishing of g makes every quotient defined.

(b) $(f + g)(1) = 2$, whereas a multiplicative function has value 1 at 1.

(c) Take $f(n) = 1$ and $g(n) = n$, which are both completely multiplicative. Then $A(n) = (n + 1)/2$. Although $A(1) = 1$,

$$A(6) = \frac{7}{2} \neq 3 = \frac{3}{2} \cdot 2 = A(2)A(3), \qquad \gcd(2, 3) = 1.$$

Grading: (a) Each proof, including its value at 1: 2. (b) Value at 1 and conclusion: 2. (c) Valid multiplicative examples: 1; normalized values: 1; coprime inputs: 1; failed equality: 1.

3. Counting and adding divisors [14 points]

Define $\tau(n) = \sum_{d|n} 1$ and $\sigma(n) = \sum_{d|n} d$. Write $n = \prod_{i=1}^{r} p_i^{a_i}$, with distinct primes p_i.

(a) Describe the divisors of n using exponents. Deduce product formulas for $\tau(n)$ and $\sigma(n)$, and explain why both functions are multiplicative. (6)

(b) Compute $\tau(540)$ and $\sigma(540)$. (4)

(c) Prove that $\tau(n)$ is odd if and only if n is a perfect square. Include $n = 1$. (4)

Solution

(a) Every divisor is uniquely $\prod_i p_i^{b_i}$ with $0 \leq b_i \leq a_i$. Counting choices, or distributing a product of finite sums, gives

$$\tau(n) = \prod_i (a_i + 1), \qquad \sigma(n) = \prod_i (1 + p_i + \cdots + p_i^{a_i}).$$

At 1 both products are empty and have value 1. Coprime inputs have disjoint prime factors, so the formulas give multiplicativity.

(b) $540 = 2^2 \cdot 3^3 \cdot 5$, hence

$$\tau(540) = 3 \cdot 4 \cdot 2 = 24, \qquad \sigma(540) = (1 + 2 + 4)(1 + 3 + 9 + 27)(1 + 5) = 1680.$$

(c) A product of integers is odd exactly when all factors are odd. Thus $\tau(n)$ is odd exactly when every $a_i + 1$ is odd, equivalently when every a_i is even. Unique factorization says this is exactly the condition for n to be a square. Also $\tau(1) = 1$ and $1 = 1^2$. Alternatively, pair divisors $d, n/d$; an unpaired divisor occurs exactly when $d^2 = n$.

Grading: (a) Unique exponent description: 2; each product formula: 1; multiplicativity including 1: 2. (b) Each calculation: 2. (c) Parity criterion: 2; square criterion and both directions: 1; case 1: 1.

II. Totients and divisor sums

4. The totient from excluded multiples [14 points]

Let $\varphi(n)$ count integers $1 \le k \le n$ with $\gcd(k, n) = 1$; in particular, $\varphi(1) = 1$.

(a) For a prime p and $a \ge 1$, prove $\varphi(p^a) = p^a - p^{a-1}$. (3)

(b) Apply inclusion–exclusion to the multiples of the distinct primes dividing n to prove

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

Use this formula to prove that φ is multiplicative. (6)

(c) Compute $\varphi(540)$. (2)

(d) Prove that $\varphi(n)$ is even for every $n > 2$. *Hint:* pair k with $n - k$. (3)

Solution

(a) An integer is not coprime to p^a exactly when p divides it. Among $1, \dots, p^a$, there are p^{a-1} such multiples. Subtract these from the p^a integers.

(b) If the distinct prime divisors are $p_1, \dots, p_r$, being coprime means avoiding every multiple of these primes. For a subset S of the primes, the number divisible by all of them is $n / \prod_{p \in S} p$; the denominator divides n. Inclusion–exclusion therefore gives

$$\varphi(n) = \sum_{S \subseteq \{p_1, \dots, p_r\}} (-1)^{|S|} \frac{n}{\prod_{p \in S} p} = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right).$$

For coprime m, n the prime sets are disjoint; the product for mn separates into the products for m and n, with leading factor $mn = m \cdot n$. Also $\varphi(1) = 1$.

(c) $\varphi(540) = 540(1 - 1/2)(1 - 1/3)(1 - 1/5) = 144$.

(d) For $n > 2$, every counted k lies in $1, \dots, n - 1$. Since $\gcd(n - k, n) = \gcd(k, n)$, the map $k \mapsto n - k$ pairs counted integers and is its own inverse. A fixed point would have $2k = n$, so $\gcd(k, n) = k$; coprimality would force $k = 1, n = 2$. This is excluded. All pairs have two distinct elements, so the count is even.

Grading: (a) Exclude exactly the multiples of p: 1; count: 1; subtraction: 1. (b) Intersection counts: 2; inclusion–exclusion and product: 2; multiplicativity including 1: 2. (c) Setup and value: 2. (d) Pair preserves coprimality: 1; no fixed points for $n > 2$: 1; even count: 1.

5. Multiplication by a divisor sum [14 points]

The *Dirichlet convolution* of arithmetic functions is

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d).$$

Let $\mathbf{1}(n) = 1$, $\mathrm{id}(n) = n$, and let $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$.

(a) Suppose $\gcd(m, n) = 1$. Prove that every divisor of mn has a unique expression ab with $a \mid m$ and $b \mid n$. (4)

(b) Use (a) to prove that $f * g$ is multiplicative when f and g are multiplicative. Check the value at 1 and explain each use of coprimality. (6)

(c) Identify $\mathbf{1} * \mathbf{1}$ and $\mathbf{1} * \mathrm{id}$ in terms of the functions already studied. For an arbitrary arithmetic function f, prove $f * \varepsilon = f$. (4)

Solution

(a) No prime divides both m and n. Assign each prime power in a divisor of mn to the factor coming from m or from n. This gives the required a, b and makes them unique. Conversely, every such product ab divides mn.

(b) Set $h = f * g$. We have $h(1) = f(1)g(1) = 1$. For coprime m, n, both $\gcd(a, b) = 1$ and $\gcd(m/a, n/b) = 1$, so

$$\begin{aligned} h(mn) &= \sum_{a \mid m} \sum_{b \mid n} f(ab) g((m/a)(n/b)) \\ &= \sum_{a \mid m} \sum_{b \mid n} f(a) f(b) g(m/a) g(n/b) \\ &= \left(\sum_{a \mid m} f(a) g(m/a) \right) \left(\sum_{b \mid n} f(b) g(n/b) \right) = h(m) h(n). \end{aligned}$$

(c) $(\mathbf{1} * \mathbf{1})(n) = \sum_{d \mid n} 1 = \tau(n)$. Also $(\mathbf{1} * \mathrm{id})(n) = \sum_{d \mid n} n/d = \sigma(n)$, since $d \mapsto n/d$ permutes the divisors. Finally, only the term $d = n$ survives in $\sum_{d \mid n} f(d)\varepsilon(n/d)$, giving $f(n)$.

Grading: (a) Existence: 2; uniqueness and converse: 2. (b) Value at 1: 1; divisor reindexing: 1; both coprimality conditions: 2; product factorization: 2. (c) τ: 1; σ: 1; identity proof: 2.

6. Möbius cancellation [14 points]

An integer is *squarefree* if no prime square divides it. Define

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & n \text{ is not squarefree.} \end{cases}$$

(a) Prove that μ is multiplicative. Show that it is not completely multiplicative. (3)

(b) Find $\sum_{d \mid p^a} \mu(d)$ for a prime p and $a \geq 1$. Use Problem 5 to prove $\mu * \mathbf{1} = \varepsilon$, including its value at 1. (5)

(c) You may use associativity and commutativity of convolution. Prove the following identities for every $n \geq 1$. (6)

$$\sum_{d \mid n} \mu(d)\sigma(n/d) = n, \qquad \sum_{d \mid n} \mu(d)\tau(n/d) = 1.$$

Solution

(a) We have $\mu(1) = 1$. If either coprime input has a prime square factor, both $\mu(mn)$ and $\mu(m)\mu(n)$ vanish. Otherwise their disjoint lists of primes concatenate, so the signs multiply. Complete multiplicativity fails since $\mu(p^2) = 0$ but $\mu(p)^2 = 1$.

(b) The prime-power sum is $\mu(1) + \mu(p) = 1 - 1 = 0$; all higher powers contribute zero. By Problem 5, $H = \mu * \mathbf{1}$ is multiplicative. Thus for $n > 1$, its product over prime-power factors contains a zero, so $H(n) = 0$. At 1, $H(1) = 1$. This is precisely ε.

(c) Problem 5 gives $\sigma = \mathbf{1} * \mathrm{id}$ and $\tau = \mathbf{1} * \mathbf{1}$. Hence

$$\mu * \sigma = (\mu * \mathbf{1}) * \mathrm{id} = \varepsilon * \mathrm{id} = \mathrm{id}, \qquad \mu * \tau = (\mu * \mathbf{1}) * \mathbf{1} = \varepsilon * \mathbf{1} = \mathbf{1}.$$

Evaluating these function identities at n gives the two requested sums. They also hold at 1.

Grading: (a) Multiplicativity with both cases and 1: 2; counterexample to complete multiplicativity: 1. (b) Prime-power sum: 2; multiplicativity and reduction to prime powers: 2; value at 1: 1. (c) Each convolution calculation and its translation to the sum: 3.

III. Two applications

7. An alternating divisor sum detects squares [12 points]

For $n = \prod_{i=1}^{r} p_i^{a_i}$ define $\Omega(n) = a_1 + \cdots + a_r$, with $\Omega(1) = 0$, and set $\lambda(n) = (-1)^{\Omega(n)}$. Thus Ω counts prime factors *with multiplicity*.

(a) Prove that λ is completely multiplicative. (4)

(b) Evaluate $\sum_{j=0}^{a} \lambda(p^j)$ and deduce, for every n,

$$\sum_{d \mid n} \lambda(d) = \begin{cases} 1, & n \text{ is a perfect square,} \\ 0, & n \text{ is not a perfect square.} \end{cases}$$

Justify the passage from prime powers to general n, and include $n = 1$. (6)

(c) Evaluate $\sum_{d \mid 360} \lambda(d)$ and $\sum_{d \mid 900} \lambda(d)$ without listing all the divisors. (2)

Solution

(a) Prime exponents add in a product whether or not its factors are coprime, so $\Omega(mn) = \Omega(m) + \Omega(n)$. Therefore $\lambda(mn) = (-1)^{\Omega(m)+\Omega(n)} = \lambda(m)\lambda(n)$ for all m, n, and $\lambda(1) = 1$.

(b) The prime-power sum is $1 - 1 + \cdots + (-1)^a$, equal to 1 for even a and 0 for odd a. By Problem 5 the divisor-sum function $\lambda * \mathbf{1}$ is multiplicative. Its value at $\prod_i p_i^{a_i}$ is the product of these alternating sums, which is 1 exactly when every a_i is even, and 0 otherwise. This is the square criterion from unique factorization. At 1, the divisor sum is $\lambda(1) = 1$.

(c) $360 = 2^3 \cdot 3^2 \cdot 5$ is not a square, so its sum is 0. Since $900 = 2^2 \cdot 3^2 \cdot 5^2 = 30^2$, its sum is 1.

Grading: (a) Exponents add without a coprimality restriction: 2; sign product and value at 1: 2. (b) Alternating sum: 2; multiplicativity and product: 2; square criterion: 1; case 1: 1. (c) Each evaluated sum: 1.

8. A sum of gcds [12 points]

Define $G(n) = \sum_{k=1}^{n} \gcd(k, n)$.

(a) Group the integers k according to $d = \gcd(k, n)$ to prove

$$G(n) = \sum_{d \mid n} d\,\varphi(n/d).$$

Explain why the group with gcd equal to d has $\varphi(n/d)$ members. (5)

(b) Use the earlier problems to deduce that G is multiplicative. (2)

(c) For a prime p and $a \geq 1$, derive

$$G(p^a) = (a+1)p^a - ap^{a-1}.$$

Then compute $G(60)$ without summing 60 gcds individually. (5)

Solution

(a) The group with gcd d consists exactly of $k = dj$ with $1 \leq j \leq n/d$ and $\gcd(j, n/d) = 1$. There are $\varphi(n/d)$ choices, each contributing d to the sum. These groups partition $1, \ldots, n$, so their contributions add to the formula. The case $d = n$ contains only $k = n$, consistent with $\varphi(1) = 1$.

(b) The formula says $G = \mathrm{id} * \varphi$; both factors are multiplicative, so Problem 5 applies. In particular $G(1) = 1$.

(c) The term $d = p^a$ contributes $p^a\varphi(1) = p^a$. For $d = p^j$ with $0 \leq j < a$, the contribution is

$$p^j\varphi(p^{a-j}) = p^j(p^{a-j} - p^{a-j-1}) = p^a - p^{a-1}.$$

There are a such terms, giving $G(p^a) = p^a + a(p^a - p^{a-1})$ as required. Finally,

$$G(60) = G(4)G(3)G(5) = (3 \cdot 4 - 2 \cdot 2)(2 \cdot 3 - 1)(2 \cdot 5 - 1) = 8 \cdot 5 \cdot 9 = 360.$$

Grading: (a) Gcd class bijection: 3; weighted contribution: 1; partition and sum: 1. (b) Convolution and multiplicativity: 2. (c) Prime-power derivation including its final term: 3; factorization and value at 60: 2.

Optional challenge: Even perfect numbers [10 extra points]

An integer N is *perfect* if its positive divisors other than N sum to N, equivalently $\sigma(N) = 2N$.

(a) Let $r \geq 2$ and suppose $2^r - 1$ is prime. Prove that $N = 2^{r-1}(2^r - 1)$ is perfect. Give the examples from $r = 2, 3, 5$. (4)

(b) Prove that every even perfect number has the form in (a). *Hint:* write $N = 2^a m$ with $a \geq 1$ and m odd, and put $q = 2^{a+1} - 1$. First show $q \mid m$. Write $m = qt$ and show $\sigma(m) = m + t$. If $t > 1$, compare this with the contribution of the three distinct divisors $1, t, m$. (6)

Solution

(a) Set $q = 2^r - 1$, an odd prime. The factors 2^{r-1} and q are coprime, so Problem 3 gives

$$\sigma(N) = (1 + 2 + \cdots + 2^{r-1})(1 + q) = (2^r - 1)2^r = 2N.$$

The examples are $6, 28, 496$.

(b) The perfect-number equation is

$$(2^{a+1} - 1)\sigma(m) = 2^{a+1}m, \quad \text{or} \quad q\sigma(m) = (q+1)m.$$

Since $\gcd(q, q+1) = 1$, Euclid's lemma gives $q \mid m$. Write $m = qt$. Dividing by q gives $\sigma(m) = (q+1)t = m+t$. We have $q \geq 3$, so $t < m$. If $t > 1$, the three distinct divisors $1, t, m$ imply $\sigma(m) \geq 1 + t + m > m + t$, a contradiction. Thus $t = 1$, $m = q > 1$, and $\sigma(m) = m + 1$. A composite m would have an additional positive divisor besides $1, m$, so m must be prime. Setting $r = a + 1 \geq 2$ gives the required form.

Grading: (a) Coprime factorization and divisor-sum formula: 2; perfect-number equality: 1; examples: 1. (b) Equation and $q \mid m$: 2; substitution and divisor contradiction: 2; $t = 1$, primality, and the final form: 2.